Physics 214/224, Practice Quiz 3

Primary instructor: Yurii Maravin

WID:__________________________________ NAME:____

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

For two quick points, circle your studio instructor:

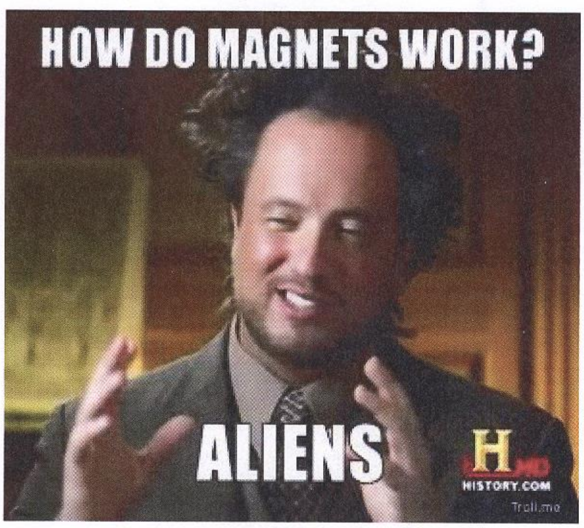
Lohman (TU 7:30) Lohman (TU 9:30) Weaver (TU 9:30)

Magrakvelidze (TU 11:30) Maravin (TU 1:30) Weaver (TU 3:30)

Magrakvelidze (WF 7:30) Magrakvelidze (WF 9:30)

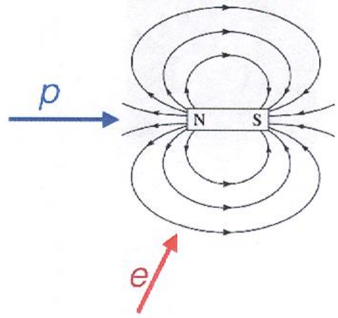
Golin (WF 11:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.



Part I: All questions to be answered on your Scantron card (48 points total)

The next two questions refer to the image below



1. (3 points) A proton moves towards the north pole of the bar magnet (blue arrow). Which way will it be deflected?

   (a) upwards
   (b) downwards
   (c) Into the page
   (d) out of the page
   (e) very little deflection will occur

2. (3 points) An electron moves towards the side of the bar magnet, as indicated by the red arrow. Which way will it be deflected?

   (a) to the left
   (b) to the right
   (c) Into the page
   (d) out of the page
   (e) very little deflection will occur

3. (3 points) Two parallel wires carrying currents I_1 and I_2 in the same direction are spaced by some distance r. What is the correct statement about the interaction forces between these two wires?

   (a) They experience a repulsive force that is inversely proportional to the distance between the wires squared and proportional to the product of the currents $I_1 \cdot I_2$.
   (b) They experience an attractive force that is inversely proportional to the distance between the wires squared and proportional to the product of the currents $I_1 \cdot I_2$.
   (c) They experience no forces whatsoever.
   (d) They experience a repulsive force that is inversely proportional to the distance between the wires and proportional to the product of the currents $I_1 \cdot I_2$.
   (e) They experience an attractive force that is inversely proportional to the distance between the wires and proportional to the product of the currents $I_1 \cdot I_2$.

$$B_{12} = \frac{\mu_0 I_1}{2\pi r} \Rightarrow F \sim B_{12} I_2 = \frac{\mu_0 I_1 I_2}{2\pi r} \cdot L$$

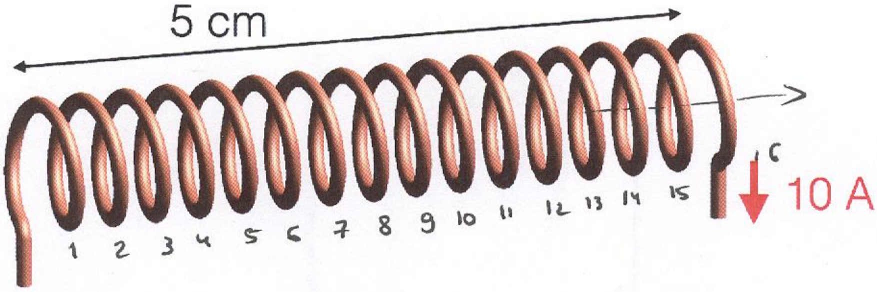
4. (3 points) An electron and a proton ($m_p/m_e = 1836$) have the same velocity and move in a constant $\vec{B}$-field at right angles to the $\vec{B}$. How will the radius of the electron circle compare to that of the proton?

(a) $R_e/R_p = 1836$
(b) $R_e/R_p = 43$
(c) $R_e/R_p = 1$
(d) $R_e/R_p = \frac{1}{1836}$

$F = qvB = \frac{mv^2}{R}$ $\quad R = \frac{mv^2}{qvB} = \frac{mv}{qB}$

$\frac{R_e}{R_p} = \frac{m_e}{m_p} = \frac{1}{1836}$

5. (3 points) A solenoid has 10 A of current flowing through it, as given in the figure below. What is the strength of the magnetic field in the middle of the solenoid?



(a) 0.0040 T
(b) 0.0020 T
(c) 0.0010 T
(d) 0.0080 T

$5\,cm \cdot B = \mu_0 \cdot 10\,A \cdot 16$

$B = \frac{160 \cdot 4\pi \cdot 10^{-7}}{5 \cdot 10^{-2}}\,T = 4.02 \cdot 10^{-3} \approx 0.004\,T$

6. (3 points) For the problem above, what is the direction of the magnetic field in the middle of solenoid?

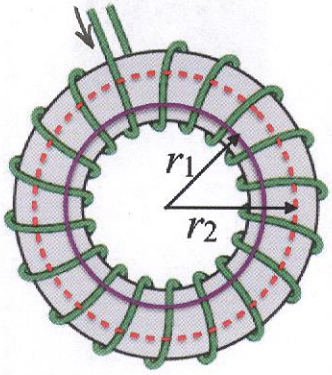
(a) to the left
(b) to the right
(c) radially outwards from the center axis of symmetry of the solenoid
(d) radially inwards towards the center axis of symmetry of the solenoid

7. (3 points) An electron travels through a vacuum with a speed of 10^6 m/s. At some point, a uniform magnetic field is created in the area of electron's trajectory with the strength of 1 Tesla. The electron travels a total of 1.2 m before the field is turned off. What is the work done by the magnetic field on the moving electron?

(a) 0 J
(b) 1.92×10^{-13} J
(c) -1.92×10^{-13} J
(d) Impossible to tell with the information given

$\vec{F} \perp \vec{v}$! $\quad (\vec{F} = q\vec{v} \times \vec{B})$

8. (3 points) How do magnitudes of the magnetic fields B_1 and B_2 in the bulk of a toroidal magnet compare at a radial distances r_1 and r_2, as illustrated in the image below? Assume some non-zero current flows in the wire that forms a toroid.



$\oint \vec{B} d\vec{l} = \mu_0 N I$

$2\pi r B = \mu_0 N I$

$B \sim \frac{1}{r}$

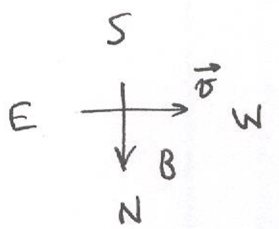
$\frac{B_1}{B_2} = \frac{r_2}{r_1}$

(a) Magnetic fields are the same, they are uniform inside a toroid
(b) $B_1/B_2 = r_1/r_2$
(c) $B_1/B_2 = r_2/r_1$
(d) $B_1/B_2 = (r_1/r_2)^2$
(e) $B_1/B_2 = (r_2/r_1)^2$

9. (3 points) In the figure for the previous problems, the current flows in the toroid from the left-most wire. What is the direction of the magnetic field in the bulk of the toroid?

(a) Clockwise
(b) Counter-clockwise
(c) Radially outwards
(d) Radially inwards

10. (3 points) Suppose a dust cloud drifting from east to west acquired a negative charge due to triboelectric effect. Which way will the magnetic force due to the Earth's magnetic field act on the cloud?



$\vec{F} = q\vec{v} \times \vec{B}$

$F \rightarrow$ up

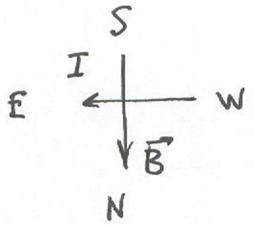
(a) Up
(b) Down
(c) South
(d) North

11. (3 points) A compass needle points north in Kansas. Which way would it point in Australia?

(a) North
(b) South
(c) East
(d) West
(e) compasses do not work down under.

12. (3 points) Estimate the magnitude of the magnetic force on a 100 meter length of a power line carrying 1 A of current in a west to east direction across Kansas. Assume the earth's B-field points due north.

(a) 5×10^{-4} N
(b) 5×10^{-3} N
(c) 5×10^{-2} N
(d) 5×10^{-1} N



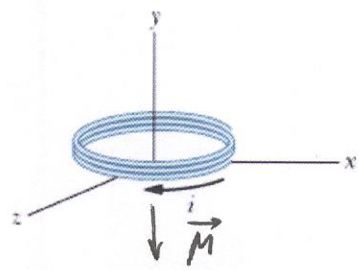
$\vec{F} = I\vec{L}\times\vec{B}$ $\qquad \frac{|\vec{F}|}{L} = IB$

$F = 100m \cdot 0.5 \cdot 10^{-4} T \cdot 1 A = 5 \cdot 10^{-3} N$

13. (3 points) We have two "field calculator" methods available: Ampere's Law and the Biot-Savart Law. Which method would be most useful for calculating fields from a single N-turn coil of wire and from a long coaxial cable?

(a) N-turn coil: Ampere's Law. Coaxial cable: Ampere's Law
(b) N-turn coil: Biot-Savart Law. Coaxial cable: Ampere's Law
(c) N-turn coil: Ampere's Law. Coaxial cable: Biot-Savart Law
(d) N-turn coil: Biot-Savart Law. Coaxial cable: Biot-Savart Law

14. (3 points) A coil shown below carries current $i = 2.00$ A in the direction indicated, is parallel to an xz plane, has 3 turns and an area of 4.00×10^{-3} m^2, and lies in a uniform magnetic field $\vec{B} = (1.00\hat{\imath} - 2.00\hat{\jmath} + 4.00\hat{k})$ mT.



$|\mu| = 2.0A \cdot 3 \cdot 4 \cdot 10^{-3} m^2 =$
$= 2.4 \cdot 10^{-2} A \cdot m^2$

The potential energy of this magnetic dipole is

(a) 48.0 μJ
(b) -48.0 μJ
(c) 16.0 μJ
(d) -16.0 μJ
(e) neither of the above-mentioned choices

$PE = -\vec{\mu}\vec{B} = -\hat{\jmath} \cdot |\mu| \cdot (1\hat{\imath} - 2\hat{\jmath} + 4\hat{k}) \cdot 10^{-3} T =$
$= 2|\mu| \cdot 10^{-3} T =$
$= 4.8 \cdot 10^{-5} T \cdot A \cdot m^2 = 48 \mu J$

15. (3 points) When an external magnetic field of 2.00 mT is applied in a material, a measured total magnetic field of 200 mT is measured with a Hall's probe. This material is

(a) is ferromagnetic
(b) is diamagnetic
(c) is paramagnetic
(d) is non-magnetic

$1 + \chi_M = 200 \Rightarrow \chi_M = 199$ ~ ferromagnetic

16. (3 points) An unknown metal causes the magnetic field produced by a solenoid to increase by a factor of 700 when the metal is inserted into the solenoid, and the metal continues to produce a magnetic field when the solenoid's current is turned off. This metal might very likely contain appreciable amounts of

(a) copper — dia

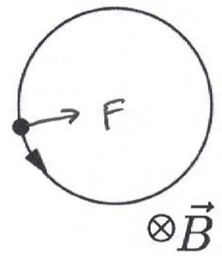
✓ (b) iron

(c) aluminum — para

(d) bismuth — dia

Part II: Work out this part in the space provided. Show your work! (50 points total)

17. (16 points) In the figure below, a charged particle is moving in a uniform magnetic field $\vec{B}$ pointing into the page. The particle is either a proton or an electron. It experiences a magnetic force of magnitude 1.6×10^{-15} N. (You might need the following info: for proton, the charge $q = +1.6\times10^{-19}$ C and mass $m_p = 1.67\times10^{-27}$ kg. The electron has a mass $m_e = 9.1\times10^{-31}$ kg.)



(a) (2 points) Is the particle a proton or an electron? Explain.

charge is positive, so proton

(b) (5 points) Determine the particle's speed

$q v B = \frac{m v^2}{R}$ $\quad v = \frac{qBR}{m}$ $\quad$ missing B or R

(c) (5 points) Determine the period of the motion

missing B

$T = \frac{2\pi R}{v} = \frac{2\pi R}{qBR} m = \frac{2\pi m}{qB} =$

(d) (4 points) Circle the correct statements below. If the other particle (to that stated in (a)) was present, for the **same** magnetic field and magnetic force, this particle's

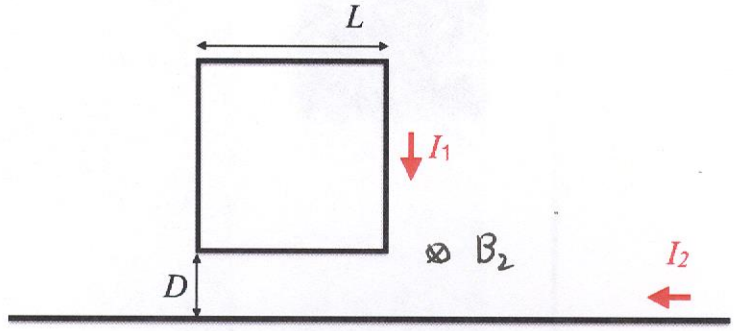
- direction of motion is clockwise/counter-clockwise
- it's speed is larger/smaller/unchanged
- it's radius of motion is larger/smaller/unchanged
- it's period is larger/smaller/unchanged

$F = qvB$
if F and B the same, v – same.

$R = \frac{mv}{qB} \Rightarrow R_e \sim m_e$ smaller

$T \sim m$ – smaller

18. (14 points) A square loop with a side size $L = 0.1$ m carries a current of $I_1 = 10$ A in a clockwise direction, as shown on picture below. The loop is positioned at a distance $D = 10$ m from a very long wire carrying a current $I_2 = 20$ A from right to left.



(a) (4 points) Calculate the magnetic field from the long wire at any point that is separated from the wire at a distance r

$$B_2(r) \cdot 2\pi r = \mu_0 I_2 \Rightarrow B_2(r) = \frac{\mu_0 I_2}{2\pi r}$$

(b) (4 points) What is the magnitude and the direction of the net force on the loop exerted by the wire?

Towards the wire

$$F = B_2(D) L I_1 - B_2(D+L) L I_1 =$$

$$= \frac{L I_1 \mu_0 I_2}{2\pi}\left(\frac{1}{D} - \frac{1}{D+L}\right) = \frac{\mu_0 L^2 I_1 I_2}{2\pi} \frac{1}{D(D+L)}$$

(c) (2 points) What is the magnetic dipole moment of the loop of wire?

$|\vec{\mu}| = I_1 L^2$ directed in the page

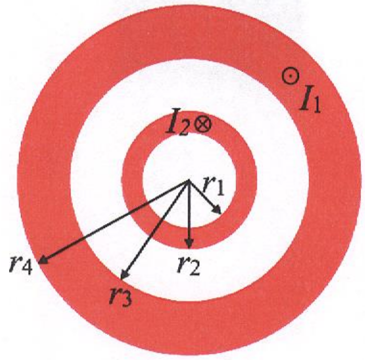
(d) (4 points) Use the formulae that you obtained in (a) and (c) to calculate a force on the loop, using the approximation $F = |\frac{dB}{dr}|\mu_B$, where μ_B is the magnetic dipole moment of the loop. How does the answer compares with what you obtained in (b)?

$$\frac{dB}{dr} = -\frac{\mu_0 I_2}{2\pi r^2} \qquad F = \left.\frac{\mu_0 I_2}{2\pi r^2}\right|_{r=D} \cdot I_1 L^2 = \frac{\mu_0 I_1 I_2 L^2}{2\pi D^2}$$

the difference between (b) and (d) is $\sim \frac{1}{D^2}$ vs $\frac{1}{D(D+L)}$

if $L \ll D$, then answers are same.

19. (10 points) There are two very long concentric conducting pipes. The inner pipe has inner and outer radii $r_1 = 10$ cm and $r_2 = 12$ cm. These for outer pipe are $r_3 = 20$ cm and $r_4 = 25$ cm, respectively. The current in the inner pipe, $I_1 = 10$ A flows into the page, and the current in the outer pipe $I_2 = 10$ A flows out of the page.



(a) (1 point) what is the magnetic field at the radius $r = 0$ cm from the center?

Zero, due to symmetry

(b) (2 points) what is the magnetic field at the radius $r = 9$ cm from the center?

$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc.}$ Use circles as Ampere's circles in this problem, so due to symmetry $\oint \vec{B} \, d\vec{\ell} = 2\pi r B = \mu_0 \cdot 0 \Rightarrow$

$B = 0\,T.$

(c) (3 points) what is the magnetic field at the radius $r = 11$ cm from the center?

Current density for inner pipe $\vec{j} = \dfrac{I_2}{\pi(r_2^2 - r_1^2)}$

$$2\pi r B = \mu_0 j \cdot (\pi r^2 - \pi r_1^2) = \mu_0 \frac{I_2(\pi r^2 - \pi r_1^2)}{\pi(r_2^2 - r_1^2)} \Rightarrow B = \frac{\mu_0 I_2}{2\pi r}\left(\frac{r^2 - r_1^2}{r_2^2 - r_1^2}\right) =$$

$= 8.68\,\mu T$

(d) (3 points) what is the magnetic field at the radius $r = 15$ cm from the center?

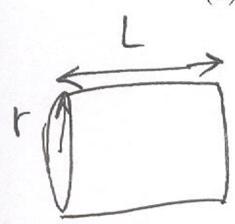
$$2\pi r B = \mu_0 I_2 \Rightarrow B = \frac{\mu_0 I_2}{2\pi r} = 13.3\,\mu T$$

(e) (1 point) what is the magnetic field at the radius $r = 50$ cm from the center?

$$2\pi r B = \mu_0 (I_2 - I_1) = 0 \Rightarrow B = 0\,T$$

20. (10 points) Suppose we wish to magnetize the huge volume of a cylinder-shaped spectrometer at the CERN Large Hadron Collider to a value of 3.8 T. We choose a solenoid to accomplish this. The solenoid is 13 m long and 3 m in radius. A physicists-r-us power supply can deliver 19500 A to the solenoid.

(a) (3 points) Show that we need to wrap a conductor around the outside of the cylinder a total of 2199 times over its full length to produce this field with the given current.



$$L \cdot B = \mu_0 N I \Rightarrow N = \frac{LB}{\mu_0 I} = \frac{13\,m \cdot 3.8\,T}{4\pi \cdot 10^{-7} \cdot 19500\,A} = 2016$$

I get 2016 turns. 2199 is a typo.

(b) (3 points) Show that we would need about 40 km of wire to achieve this

$$L = 2016 \cdot 2\pi r = 2016 \cdot 2\pi \cdot 3\,m \simeq 38 \cdot 10^3\,m \simeq 38\,km$$

(c) (4 points) Suppose the wire had a resistance of 1 Ω/km. How much power would be required to operate the solenoid in this case? Does the value you get seem practically achievable? For reference, the Wolf Creek nuclear plant in Burlington, KS can produce electrical power at a rate of 1250 MegaWatts.

$$R = 1\frac{\Omega}{km} \cdot 38\,km = 38\,\Omega$$

$$P = I^2 R = (19500\,A)^2 \cdot 38\,\Omega = 14 \cdot 10^9\,W = 14{,}000\,MW$$

It will take ~11 nuclear plants to run the magnet!